



Digital Signal Intelligence

A New Information Substrate for Insurance White paper

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Abstract:

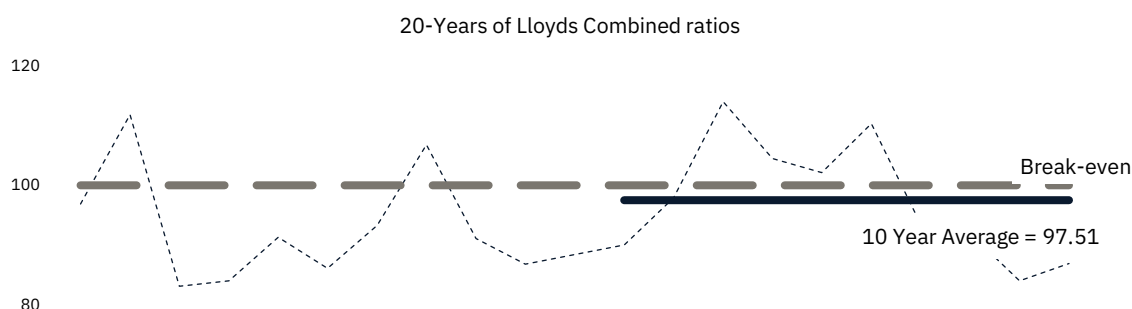
Digital Signal Intelligence (DSI) defines a new, externally observable information substrate for Insurance, enabling deterministic, autonomous risk assessment, exposure estimation, loss analysis, and pricing at scale.

The Automation Myth

Introduction

For two decades, the insurance industry has attempted to automate the traditional underwriting process. Vendors, insurtechs, and internal transformation programmes have all pursued the same objective: extract structured data from unstructured submissions, feed it into rating models, and accelerate underwriting decisions.

Yet twenty years of Lloyd's combined ratios show the same truth as every major market: insurance is flat-lining. Despite billions invested, the results have been marginal. Zero-touch underwriting remains out-of-reach, expense ratios remain structurally high, and underwriting workflows remain dependent on manual intervention.



The core assumption behind these efforts is that traditional underwriting inputs can be automated.

This assumption is incorrect.

[The inputs themselves are the constraint.](#)

The Information Substrate Problem

Traditional inputs exist in infinite formats. Broker spreadsheets, PDFs, emails, and supplemental questionnaires have no canonical schema and no deterministic structure. No extraction system, including modern or future AI models, can consistently transform these artefacts into the reliable, field-level inputs required to facilitate zero-touch underwriting.

A system built on these inputs cannot achieve deterministic automation.

[To enable autonomous underwriting, the information substrate must change.](#)

Digital Signal Intelligence

Digital Signal Intelligence (DSI) replaces self-reported inputs with externally observable digital signals that describe how an organisation behaves, how it operates, and how it is connected to the digital ecosystem. These signals are structured, verifiable, and machine-readable by design.

[DSI provides a deterministic foundation for underwriting, enabling automated risk assessment, exposure estimation, and pricing at scale.](#)

DSI is not experimental. It is based on the same network-based principles that underpin modern search and recommendation systems, beginning with Google's PageRank in 1998.

This paper defines the DSI methodology, the principles that govern it, the architecture that implements it, and the economic impact it delivers.

The Core Thesis

Insurance underwriting requires understanding two properties of an organisation:

1. How well it is managed
2. How it behaves over time

Traditional underwriting attempts to infer these properties from self-reported documents, which are subjective, inconsistent, and non-verifiable.

DSI replaces them with observable digital signals.

All organisations operate within a digital ecosystem. Their participation in networks of clients, suppliers, partners, regulators, and certification bodies creates a citation network revealing trust and standing. Their digital trail of operational discipline provides a reliable proxy for maturity. Their hiring activity and investor communications reveal financial performance. Even the absence of expected signals is meaningful because well-run companies are predictable and uniform.

DSI uses these observable signals to create a deterministic, machine-readable foundation for underwriting without requiring self-reported input.

Key Terminology

- **Coverage.**
A coverage is a defined insurance product (e.g., Cyber, D&O, Marine Hull). Each coverage contains one or more configurations.
- **Configuration.**
A configuration is the complete specification of a model for a given coverage. It defines: the signals used, the scoring algorithms, the weightings, the tier thresholds, the pricing logic, the decision rules. A configuration is the single source of truth for how a model operates.
- **Model Cycle.**
A model cycle is a complete execution of a configuration against a submission. Each cycle produces: a full signal extraction, a composite score, a risk tier, (optionally) an exposure band and complexity category, a pricing output, a decision. Every cycle is persisted as a Model Version, ensuring full auditability.
- **Signal.**
A measurable, externally observable data point used in scoring. Signals are normalised to a 0–100 scale.
- **Composite Score.**
A weighted aggregation of signals normalised to a 0–1000 scale. Composite scores map directly to risk tiers.
- **Risk Tier.**
A discrete classification (typically Tier 1–5) representing risk quality. Tiers drive pricing and decision logic.
- **Exposure Band.**
A classification of exposure magnitude (micro → very large), inferred from observable signals.
- **Complexity Category.**
A classification of exposure complexity (simple → highly complex), inferred from observable signals.

- **Deterministic Adjustments.**
Explicit, rule-based modifiers applied after tier assignment. No subjective adjustments are permitted.

Foundational Principles

To ensure DSI operates consistently and autonomously, ten mandatory principles define the methodology:

- **External Observability.**
Primary signals must be obtainable without cooperation from the insured.
- **Machine Readability.**
All signals must be analysable by automated systems without human interpretation.
- **Network Authority.**
Relationship patterns reveal quality. Models incorporate inbound authority, outbound selectivity, centrality, co-citation, and certification relationships.
- **Behavioural Inference Over Self-Reporting.**
Observable behaviour is weighted more heavily than stated intentions.
- **Absence as Signal.**
Absence of expected signals is a measurable indicator of risk.
- **Structured Data Utilisation.**
Authoritative structured data—credit ratings, ESG scores, regulatory filings—is incorporated directly.
- **Minimal Direct Inquiry.**
Direct questions must be optional and constrained to 5–10 binary or categorical questions per line of business.
- **Organisational-Level Assessment.**
Risk is assessed at the organisational level; parent behaviour is a proxy for subsidiaries and assets.
- **Simplicity in Scoring.**
Models follow a clear scoring flow: Signal → Score → Composite → Tier → Price. Composite scores always fall between 0–1000.
- **Agentic Readiness.**
Models must be executable by autonomous systems with deterministic scoring and defined decision logic.

Signal Taxonomy

DSI signals fall into seven canonical categories. All models must identify which categories they employ and document the specific signals within each.

The seven categories are:

1. **Network Authority Signals.**
Signals derived from an entity's position within networks of trust, partnership, and transaction.
2. **Technical Infrastructure Signals.**
Signals derived from observable technical implementation (TLS, DNS, exposed services, software currency, hosting infrastructure).

3. Corporate Digital Footprint Signals.
Signals derived from an entity's own digital presence (website structure, content recency, policy documentation, leadership visibility, hiring activity).
4. Behavioural Signals.
Signals derived from observable operational behaviour over time (change patterns, incident response, regulatory engagement, customer/employee interaction).
5. Public Record Signals.
Signals derived from regulatory filings, litigation records, corporate registries, IP filings, and breach disclosures.
6. Structured Data Feed Signals.
Signals derived from authoritative third-party providers (credit ratings, ESG scores, market data, classification society records).
7. Direct Inquiry Signals (Optional).
Minimal, structured questions used only when externally observable signals are insufficient.

Scoring Architecture

Models convert raw signals into deterministic underwriting outputs.

First, each signal is normalised to a value between 0–100 using defined scoring algorithms. A confidence score reflects signal availability and determines whether intervention is appropriate.

Second, signals are weighted, based on impact analysis and defined by the applicable model, and combined into a composite score on a 0 – 1000 scale.

Third, composite scores map to model configurable risk tiers. For example, Tier 1: 800 – 1000. Each tier carries a defined decision logic. For example, Tier 1: auto-approve. This ensures that risks are treated correctly and flagged for intervention when appropriate.

Fourth, risk tiers determine base-rate premium. These are calculated either directly, for example Tier 1: \$10,000, or as a % of a known value, for example Tier 1: 0.08% of TIV.

Finally, applicable deterministic adjustments are applied.

Exposure Shadow Layer

The Exposure Shadow Layer extends DSI from assessing risk quality to estimating risk size. Traditional exposure estimation relies on self-reported schedules, statements of value, and bordereaux. These artefacts are slow, inconsistent, and incompatible with automation.

DSI introduces a different approach: exposure magnitude and complexity can be inferred from the same observable signals that reveal operational maturity.

Why It Works

Organisations leave measurable digital traces of their scale:

- the breadth of their digital footprint
- the depth of their infrastructure
- the number of locations implied by regulatory filings
- the size of their partner and vendor networks
- the complexity of their technology stack

- the geographic dispersion of their operations

These signals correlate strongly with exposure magnitude.

They do not require self-reporting.

What It Adds

The Exposure Shadow Layer provides two additional outputs:

1. Exposure Band (micro → very large)
2. Complexity Category (simple → highly complex)

These operate independently of risk quality.

This separation is essential: A large, well-managed organisation is not priced like a small, poorly managed one. Conflating size and quality produces actuarially unsound pricing.

How It Integrates

The Exposure Shadow Layer augments the Scoring Architecture:

Risk Tier × Exposure Band × Complexity Category → Rate → Premium

This enables:

- two-dimensional pricing
- higher straight-through processing
- automated triage
- reduced reliance on client-provided schedules

Loss Pattern Inference

Loss data is sparse, inconsistent, and often unavailable at the required granularity. As such in the traditional model the gap between technical pricing and loss is absolute.

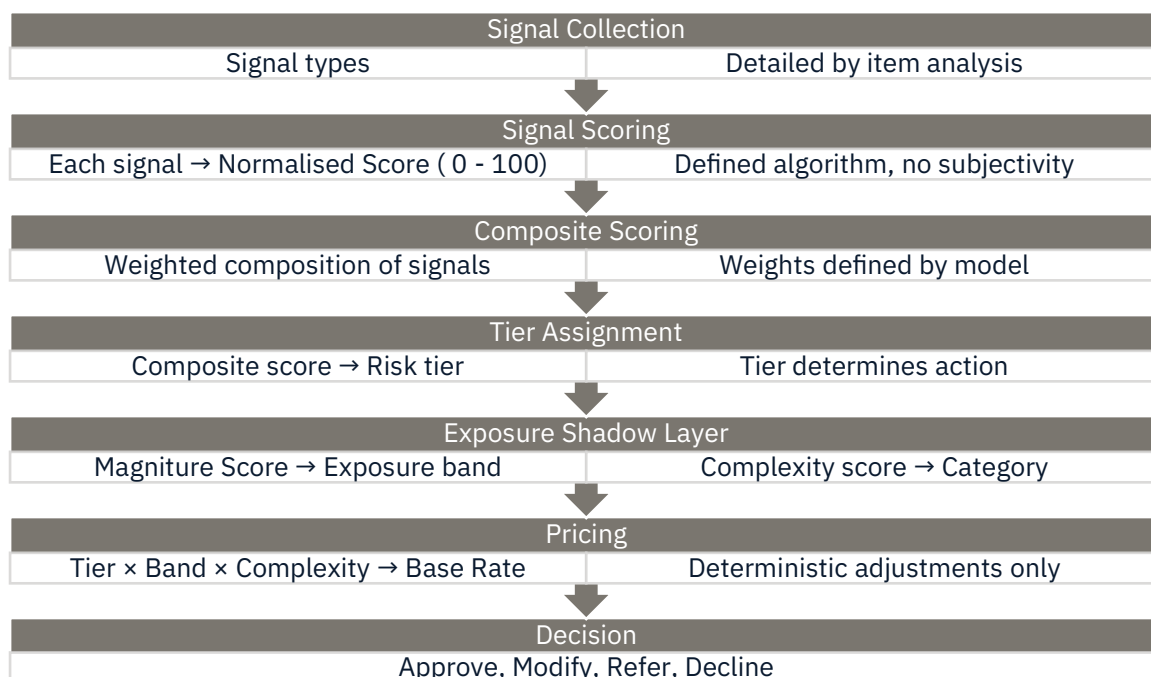
DSI signals capture behavioural and operational patterns that precede loss events, enabling inference of loss propensity even when historical loss data is absent.

This will enable:

- early detection of deteriorating risks
- segmentation within identical industry codes
- identification of portfolio correlations
- real-time monitoring and recalibration

End-to-End Model Architecture

The model architecture follows a deterministic, auditable flow. Each stage transforms observable signals into underwriting decisions.



This architecture is: deterministic, auditable, machine-readable, agentic-ready, and globally scalable.

Economic Impact

Expense Ratios

DSI targets a structural 22-point reduction in expense ratio:

Metric	Traditional Model	DSI Target
Staff Costs	12%	4%
Office and Infrastructure	8%	2%
Administration	4%	1%
Brokerage and Commissions	11%	6%

Loss Ratios

DSI targets a structural 6–12-point decrease in loss ratios:

Item	DSI Target
Loss analysis (Behaviour-Driven Loss Propensity)	4% - 8%
Continuous Model Tuning (Dynamic Calibration)	2% - 4%

Overall Portfolio Impact

On a \$500M premium book, this represents \$130M–\$170M in improved underwriting profit.

Conclusion

Digital Signal Intelligence replaces the traditional, self-reported information substrate of insurance with externally observable, machine-readable signals. This enables deterministic scoring, autonomous underwriting, continuous monitoring, and global scalability.

Insurers are publicly committing to reducing operational complexity and headcount. However, no traditional underwriting model can achieve this at scale. The bottleneck is the information substrate itself.

DSI is a structural replacement for the information substrate of underwriting, providing a viable path to radical operational efficiency and increased profitability.

Traditional Model		DSI Model	
Self-reported inputs		Externally observable signals	
Unstructured artefacts		Machine-readable data	
Manual interpretation		Deterministic scoring	
Slow, multi-step workflows		Instant assessment	
High cost per submission		Low cost per submission	
Limited scalability		Global scalability	

As digital signals expand and calibration datasets grow, DSI becomes not only a new methodology for underwriting, but the foundation upon which the next generation of insurance carriers will be built.